

Studio 8

Practical Test 2 Preparation

Objectives

To practice all of the concepts learnt so far in the paper in preparation for Practical Test 2.

Exercise 1 (Studio Handin Exercise)

Nilesh has been working on the problem for Practical Test 2 and had completed the code but decided to make some changes while having had a bit too much of the amber liquid and he's mucked up the code (pro tip: don't code while drunk!). He's still got a massive hangover so he has asked you to complete the code for him and then he'll do a similar problem for the test.

1. Download the template sketch from Moodle and open it in Processing.
2. Have a look at the parts of the code that are left and compare it against the class diagram.

Implement the abstract `GardenObject` class using the notes below:

The constructor will initialise the object using the values passed to the method. The `isSelected` field should be set to false. The `selectedColor` field should be set to a reddish colour.

The `drawObject()` and `isClicked()` methods are abstract methods.

The `deselectObject()` will set the `isSelected` field to false.

The `moveObject()` method will change the position of the object to the `x` and `y` values passed to the method.

3. Finish implementing the Plant class using the notes below:

The constructor will initialise the object using the values passed in.

The drawObject() method will make sure that ellipse mode is set to centred mode. Then it will set the fill colour to the plant's colour if it is not selected, otherwise set the fill colour to the selected colour of the object. Then it will draw a circle for the plant.

Luckily Nilesch didn't stuff up the isClicked() method so don't change it's code.

4. Finish implementing the Bench class using the notes below:

The constructor will initialise the object using the values passed in.

The drawObject() method will make sure that rect mode is set to centred mode. Then it will set the fill colour to the bench's colour if it is not selected, otherwise set the fill colour to the selected colour of the object. Then it will draw a rectangle for the bench.

Luckily Nilesch didn't stuff up the isClicked() method so don't change it's code.

5. Implement the Garden class using the notes below:

The constructor will create a new arraylist object. The selectedObject field should be set to null.

The addPlant() and addBench() methods will add the appropriate object to the arraylist.

The drawGarden() method will draw all garden objects to the sketch window.

Test your code and make sure you can add plants and objects to the garden using the appropriate key press and they are drawn correctly (don't worry about handling a click or growing the garden yet).

6. Extend the Garden class and implement the handleClick() method. This method will allow us to select an object by clicking on it and then move it to a different location by clicking somewhere else in the sketch window.

Use the following pseudo-code (talk to the demos to make sure you understand the logic of the pseudo-code):

```
IF an object has not been selected THEN
  FOR each object in the garden
    IF current object has been clicked on THEN
      Make the selected object the current object
    ENDIF
  ENDFOR
ELSE
  Move the selected object to the new position
  Deselect the selected object
  Make the selected object null
ENDIF
```

Test your code by uncommenting the handleClick code in the sketch tab and make sure you can select and move objects around.

7. Extend the Garden class and implement the growGarden() method. The method should make only plants in the garden grow.

Hint: Loop through all of the objects in the garden and check if the current object is an instance of a plant, if it is then make it grow. Research the instanceof operator in java.

Test your code and make your plants grow!

Exercise 2

You will be extending the spreadsheet problem from a previous studio. This problem was used as a Practical Test 2 problem in previous years.

1. Download the template sketch from Moodle and open it in Processing.
2. Compare the given code against the class diagram and you will see that the AbstractCell, TextCell and NumericCell classes have already been implemented and don't require any further changes.

3. Implement the Spreadsheet class using the notes below:

The CELL_WIDTH constant should be set to 300.

The CELL_HEIGHT constant should be set to 100.

The INITIAL_TEXT_SIZE constant should be set to 50.

The constructor should set the number of columns and rows to the values passed in and create a new ArrayList object.

The checkPosition() method is a private method used internally in the class only. It will check that the column number and row number passed it is a valid row and column for the spreadsheet. If both values are valid the method will return true, otherwise it will return false.

The addTextCell() and addNumericCell() methods will first check that the row and position values passed in are valid using the checkPosition() method and if they are then add the appropriate object to the list. If any value is not valid then display an error message in the console window.

Note: when adding the appropriate cell you must work out the correct x and y position from the column and row value. To work out the correct position just multiply the column or row value by the width or height of a cell respectively.

The drawSheet() method will draw all the cells in the spreadsheet.

The changeTextSize() method will change the size of the text in all cells by the amount passed in.

Test your spread sheet class by uncommenting the testing code in the sketch tab and making sure you can see the cells and the error messages for the invalid cells.

The comment out the testing code again which adds the cells to the spread sheet as you will now get the data from a csv file.

4. Extend the Spreadsheet class and implement the loadFile() method. It will process the csv file using the filename passed to the method and add the appropriate cells to the array list as the data is processed. Make sure you use the addTextCell() and addNumericCell() methods to add to the list so that the checking code also gets executed.

Test your sketch and make sure you can load the csv file.